

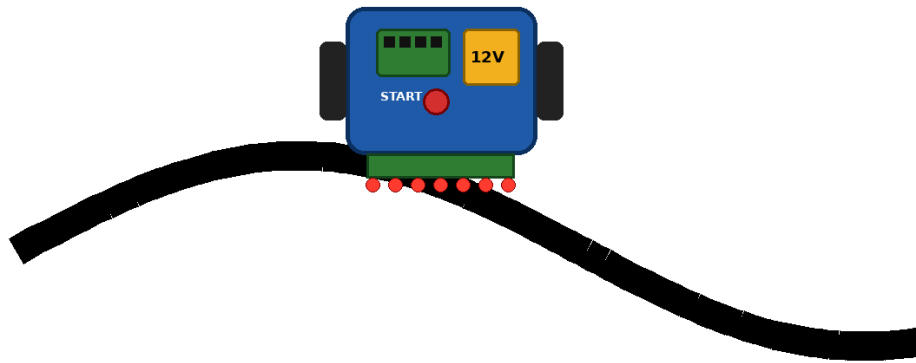
# LINE FOLLOWER ROBOT COMPETITION

## *Official Rule Book*

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### Line Follower Robot on Track

Black line | White surface | IR sensor array



## 1. Introduction

The Line Follower Robot Competition challenges autonomous robots to follow a black line on a white surface in the shortest time. The competition will test accuracy, speed, and robustness.

## 2. Team Eligibility

- 2.1 A team can consist of 2-4 members.
- 2.2 All members must be registered students.
- 2.3 One robot per team is allowed. The same robot cannot be used by two teams.
- 2.4 Team members can only be part of one team.

## 3. Robot Specifications

- 3.1 The robot must be fully autonomous. No remote control, Bluetooth, or manual control is allowed.
- 3.2 Robot dimensions: maximum 1.5 ft by 1.5 ft, with a maximum height of 0.5 ft.
- 3.3 Power supply must be onboard. Maximum 12V battery allowed.
- 3.4 Robot must have an onboard start/stop button.

## 4. Track Specifications

- 4.1 Track surface: white flex / matte white background with a black line.
- 4.2 Line width: 2.5 cm (1 inch)  $\pm$  0.5 cm.
- 4.3 The track will contain straight lines, curves, 90-degree turns, U-turns, and cross intersections. There will be NO broken lines or acute angles less than 90 degrees.
- 4.4 Start line and Finish line will be clearly marked.
- 4.5 Ambient lighting may vary. Teams must calibrate sensors accordingly.

## 5. Competition Format (3 Rounds)

### **Round 1: Basic Testing Round (Qualifying Round)**

- 5.1** Purpose: to check if the robot is functional and can follow a line.
- 5.2** Each team will get a simple straight + curve test track.
- 5.3** Each team gets 3 attempts and must complete the track in at least 1 attempt.
- 5.4** Time limit: 3 minutes per attempt.
- 5.5** Teams who complete the track will qualify for Round 2. There is no time ranking in this round.

### **Round 2: Speed Round (Time Trial)**

- 5.6** This is the main time-trial round on the full main track.
- 5.7** Each team gets 2 official runs. The best time of the 2 runs will be considered.
- 5.8** Time starts when the robot crosses the START line and stops when it crosses the FINISH line.
- 5.9** Maximum time limit per run: 5 minutes. If the robot does not finish in 5 minutes, the run is DNF (Did Not Finish).
- 5.10** The top 8 teams with the shortest times will qualify for Round 3.

### **Round 3: Final Showdown (Top Teams Battle)**

- 5.11** All top 8 teams will race SIMULTANEOUSLY on the same track.
- 5.12** Robots will start with a gap of 50 cm from each other at the start line.
- 5.13** The first robot to complete 1 full lap wins. 2nd and 3rd positions will be decided similarly.
- 5.14** If a robot goes off-track or collides and stops, it will be eliminated from that race.
- 5.15** Deliberate pushing or blocking of other robots is strictly prohibited.

## **6. Rules & Penalties**

- 6.1** Only 1 team member can be near the track during the run for start/stop.
- 6.2** No touching the robot after it starts unless permitted by the referee.
- 6.3** If the robot goes off the line for more than 3 seconds, that attempt will be considered failed.
- 6.4** Teams are allowed to re-calibrate for 2 minutes before their run.
- 6.5** The referee's decision is final.

## **7. Scoring**

- 7.1** Round 1: Pass / Fail only.
- 7.2** Round 2: Ranking based on shortest completion time.
- 7.3** Round 3: Ranking based on finishing order.
- 7.4** In case of a tie, a re-run will be conducted.

## **8. Prizes & Certificates**

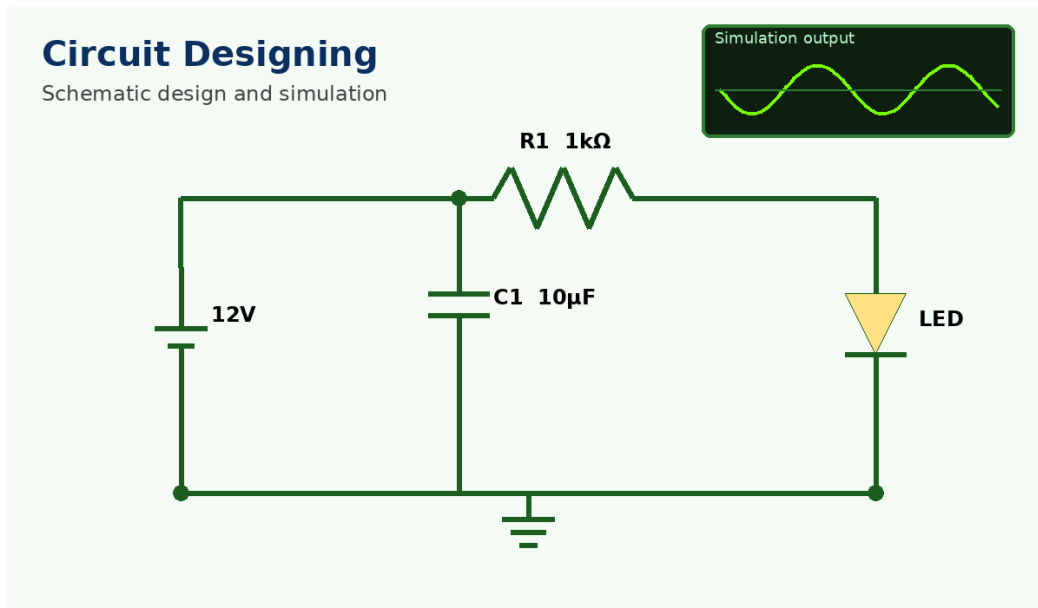
- 8.1** The winner will receive a trophy, cash prize, and certificate (cash prize will be declared later).
- 8.2** All participants will receive a participation certificate.

## **9. General Conduct**

- 9.1** Any misbehavior or argument with organizers/referees will lead to disqualification.
- 9.2** Teams must report 15 minutes before their slot.

# CIRCUIT DESIGNING COMPETITION

## Official Rules, Event Structure, and Guidelines Document



### General Regulations

#### 1. Reporting & Attendance

All participants must be present at the venue before 9:30 AM. The first round will begin sharp at 9:30 AM, and no round will be delayed due to late arrival.

#### 2. No Cheating & Integrity

Any form of unauthorized assistance, external help, or cheating during any stage of the competition will result in immediate disqualification.

#### 3. Fixed Time Limits

Each round has a strict, fixed duration. No extra time will be provided once the allotted competition time has officially ended.

#### 4. Organizer's Decision

The decision of the organizers and judges regarding evaluation, functionality validation, time-logging, and disqualification will be final and binding.

### Round 1: Screening

#### 5. Round 1 – Paper-Based MCQ Screening Written Test

A physical, written technical test covering fundamental concepts in Digital Logic Design (DLD), Electronics, and general circuit theory.

**Progression:** The top-performing teams will qualify to advance to Round 2.

### Round 2: Simulation & Debugging

#### 6. Round 2 – Simulation & Debugging (30 Minutes)

A 30-minute timed round where qualified teams perform schematic design or circuit debugging using simulation software.

**Requirements & Resources:** Each team must bring a fully charged laptop with the required simulation software installed and tested beforehand. Participants must arrange their own internet source/hotspot if needed.

**Progression:** The top-performing teams from this round will advance to the Grand Finale, strictly capped at a maximum of 5 finalist teams.

### Round 3: Grand Finale & Prizes

#### 7. Round 3 – Hardware Implementation / Grand Finale (30 Minutes)

The final pool of top teams will compete in a physical hardware building and testing challenge on breadboards/components.

**Winning Condition:** The first team to successfully build, test, and demonstrate a fully working, error-free circuit within the 30-minute limit will be declared the Winner.

**Accuracy Rule:** A rapid submission that fails functionality checks will incur penalties or be ranked behind a correctly functioning circuit.

#### 8. Podium & Awards

The competition will officially award One Winner and Two Runners-Up based on performance and completion speed in the final hardware round.

*Good luck to all participating teams! May the best engineers win.*